AI 101 for Mechanical Engineers · Module 6

A model and its learning algorithm

Lesson 6.1

What you will learn

> Distinguish a model from its learning algorithm and explain what changes during training, inference and retraining.

Lesson overview

Module 6 · Lesson 1

A model and its learning algorithm

AI 101 for Mechanical Engineers

1 A model maps inputs to outputs

Input x → Model → Prediction $\hat{y}$

$$\hat{y} = b_0 + b_1 x$$

x	input
$\hat{y}$	predicted output
b_0	intercept
b_1	slope

A learning algorithm fits the parameters. Least squares chooses the coefficients that make the sum of squared residuals as small as possible.

Residual = measured output − prediction

2 Training fits; inference applies

1 Training
Use examples to fit the intercept and slope.

2 Inference
Apply the fitted model to an input. Its coefficients stay fixed.

3 Retraining
Fit again as a separate operation, with its own data and validation.

```
model.fit(X_train, y_train)    # learn
predictions = model.predict(X_new)  # apply
```

Another prediction does not retrain the model.

3 Worked linear example

$$\hat{y} = 2 + 3x$$

Intercept = 2 | **Slope = 3**

At $x = 4$:

$$\hat{y} = 2 + 3 \times 4 = 14$$

Inference changes the input and output, not the fitted coefficients.

This arithmetic example has no stated physical units.

For a physical model: intercept unit = output unit; slope unit = output unit / input unit.

Key takeaway — **The model describes the relationship. The learning algorithm determines its fitted values.**

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A model maps inputs to outputs

$$\hat{y} = b_0 + b_1 x$$

- x: input; $\hat{y}$: predicted output
- b_0: intercept; b_1: slope
- Describes the mapping, not how coefficients were obtained

A learning algorithm obtains the parameters

- A procedure that obtains or updates parameters from data [2]
- Least squares minimizes the sum of squared residuals
- Residual: difference between a measured output and its prediction
- Related model structures can use different fitting procedures

Training fits; inference applies

Training (fitting)

- Uses examples to determine parameter values
- Linear model: fitted intercept and slope

Inference

- Applies the fitted mapping to an input
- Coefficients do not change automatically

Retraining is a separate fitting operation with its own data and validation requirements.

Inference at x = 4 gives 14

$$\hat{y} = 2 + 3x$$

$$\hat{y} = 2 + 3 \times 4 = 14$$

- Intercept 2 and slope 3 stay fixed during prediction
- Changing x changes the prediction
- Fitting again may change the intercept or slope

Coefficients carry meaning and units

Coefficient	Meaning	Unit
Intercept b_0	Predicted output at $x = 0$	Output unit
Slope b_1	Predicted change per unit change in x	Output units per input unit

> Physical meaning depends on the measurement range and model assumptions.

Do not read 14 as an engineering quantity

- This arithmetic example has no physical quantity or SI unit
- In a physical application, specify input and output units
- Module 7 works through these units in detail

fit learns parameters; predict only applies them

model.fit(X_{train}, y_{train})

- Learns parameters from the training examples

model.predict(X_{new})

- Applies the fitted model to new inputs
- Does not retrain the model

CHECK YOUR UNDERSTANDING

In the example at x = 4, which operation produces 14, and which values remain fixed?

Answer before moving on.

CHECK YOUR UNDERSTANDING

In the example at x = 4, which operation produces 14, and which values remain fixed?

> Inference produces the output 14. The fitted intercept 2 and slope 3 remain fixed during that prediction.

CHECK YOUR UNDERSTANDING

Does generating another prediction automatically retrain the model?

Answer before moving on.

CHECK YOUR UNDERSTANDING

Does generating another prediction automatically retrain the model?

> No. Retraining is a separate parameter-fitting operation that requires data and validation.

Key takeaways

- A model maps inputs to outputs; a learning algorithm fits parameters
- Training determines fitted values
- Inference applies those values to inputs
- Retraining requires a separately evaluated fit

References

[2] K. P. Murphy, Probabilistic Machine Learning: An Introduction. Cambridge, MA, USA: MIT Press, 2022.

Numbers match the course reference list. The written lesson is the authoritative text.